

Aberrant dynamic functional architecture in major depressive disorder: Vertex-Wise large-sample fMRI analyses reveal network-specific alterations and symptom associations

Received: 26 December 2024

Revised: 8 December 2025

Accepted: 20 January 2026

Cite this article as: Li, X.-Y., Lu, B., Chen, X. *et al.* Aberrant dynamic functional architecture in major depressive disorder: Vertex-Wise large-sample fMRI analyses reveal network-specific alterations and symptom associations. *Transl Psychiatry* (2026). <https://doi.org/10.1038/s41398-026-03812-1>

Xue-Ying Li, Bin Lu, Xiao Chen, Zi-Han Wang, Yu-Wei Wang, Yi-Fan Liao, Zheng-Jia-Yi Hu, Chen-Nan Wu, Han-Lin Wang, Qing-Lin Gao, Yan-Rong Chen, Si-Cheng Chen, Xin-Yu Wang, Hai-Long Liu, Ze-Di Lin, Li-Ping Cao, Guan-Mao Chen, Jian-Shan Chen, Tao Chen, Tao-Lin Chen, Yu-Qi Cheng, Zhao-Song Chu, Xi-Long Cui, Qi-Yong Gong, Wen-Bin Guo, Can-Can He, Qian Huang, Xin-Lei Ji, Feng-Nan Jia, Li Kuang, Bao-Juan Li, Feng Li, Hui-Xian Li, Tao Li, Xiao-Yun Liu, Yan-Song Liu, Zhe-Ning Liu, Yi-Cheng Long, Jian-Ping Lu, Jiang Qiu, Xiao-Xiao Shan, Tian-Mei Si, Peng-Feng Sun, Chuan-Yue Wang, Hua-Ning Wang, Xiang Wang, Ying Wang, Xin-Ran Wu, Yan-Kun Wu, Chun-Ming Xie, Guang-Rong Xie, Peng Xie, Xiu-Feng Xu, Zhen-Peng Xue, Hong Yang, Jian Yang, Hua Yu, Yong-Qiang Yu, Min-Lan Yuan, Yong-Gui Yuan, Ai-Xia Zhang, Ke-Rang Zhang, Wei Zhang, Jing-Ping Zhao, Jia-Jia Zhu, Xi-Nian Zuo, Chao-Gan Yan, Xiao-Ping Wu & on behalf of the DIRECT Consortium

We are providing an unedited version of this manuscript to give early access to its findings. Before final publication, the manuscript will undergo further editing. Please note there may be errors present which affect the content, and all legal disclaimers apply.

If this paper is publishing under a Transparent Peer Review model then Peer Review reports will publish with the final article.

Aberrant Dynamic Functional Architecture in Major Depressive Disorder: Vertex-Wise Large-Sample fMRI Analyses Reveal Network-Specific Alterations and Symptom Associations

Xue-Ying Li, Ph.D.^{1,2,3}, Bin Lu, Ph.D.^{2,3,4}, Xiao Chen, Ph.D.^{2,3,5,6}, Zi-Han Wang, M.S.^{2,3}, Yu-Wei Wang, Ph.D.⁷, Yi-Fan Liao, M.S.^{2,3}, Zheng-Jia-Yi Hu, M.S.^{2,8,9}, Chen-Nan Wu, B.Sc.^{2,3}, Han-Lin Wang, B.Sc.^{2,8,9,10,11}, Qing-Lin Gao, M.S.^{2,3}, Yan-Rong Chen, M.S.^{5,6}, Si-Cheng Chen, M.S.^{5,6}, Xin-Yu Wang, M.S.^{5,6}, Hai-Long Liu, B.Sc.^{2,3}, Ze-Di Lin, M.S.^{1,12}, Li-Ping Cao, M.D.¹³, Guan-Mao Chen, M.D.¹⁴, Jian-Shan Chen, M.D.¹³, Tao Chen, M.D.¹⁵, Tao-Lin Chen, M.D.^{16,17}, Yu-Qi Cheng, M.D.¹⁸, Zhao-Song Chu, M.D.¹⁸, Xi-Long Cui, M.D.¹⁹, Qi-Yong Gong, M.D.^{16,17}, Wen-Bin Guo, M.D.¹⁹, Can-Can He, M.D.²⁰, Qian Huang, M.D.²¹, Xin-Lei Ji, M.D.¹⁹, Feng-Nan Jia, M.D.²², Li Kuang, M.D.²¹, Bao-Juan Li, M.D.²³, Feng Li, M.D.²⁴, Hui-Xian Li, Ph.D.²⁵, Tao Li, M.D.^{26,27}, Xiao-Yun Liu, M.D.²⁸, Yan-Song Liu, M.D.²², Zhe-Ning Liu, M.D.¹⁹, Yi-Cheng Long, M.D.¹⁹, Jian-Ping Lu, M.D.²⁹, Jiang Qiu, Ph.D.³⁰, Xiao-Xiao Shan, M.D.¹⁹, Tian-Mei Si, M.D.³¹, Peng-Feng Sun, M.D.³², Chuan-Yue Wang, M.D.²⁴, Hua-Ning Wang, M.D.²³, Xiang Wang, M.D.¹⁹, Ying Wang, M.D.¹⁴, Xin-Ran Wu, M.D.³⁰, Yan-Kun Wu, M.S.³¹, Chun-Ming Xie, M.D.²⁰, Guang-Rong Xie, M.D.¹⁹, Peng Xie, M.D.^{33,34,35}, Xiu-Feng Xu, M.D.¹⁸, Zhen-Peng Xue, M.D.²⁹, Hong Yang, M.D.¹⁵, Jian Yang, M.D.³⁴, Hua Yu, M.D.^{26,27}, Yong-Qiang Yu, M.D.^{36,37,38}, Min-Lan Yuan, M.D.³⁹, Yong-Gui Yuan, M.D.²⁸, Ai-Xia Zhang, M.D.⁴⁰, Ke-Rang Zhang, M.D.⁴⁰, Wei Zhang, M.D.³⁹, Jing-Ping Zhao, M.D.¹⁹, Jia-Jia Zhu, M.S.^{36,37,38}, Xi-Nian Zuo, Ph.D.^{41,42}, Chao-Gan Yan, Ph.D.^{1,2,3,5,6,8,9,#} & Xiao-Ping Wu, M.D.^{32,#} on behalf of the DIRECT Consortium*

¹Department of Psychological and Cognitive Sciences, Tsinghua University, Beijing 100084, China

²CAS Key Laboratory of Behavioral Science, Institute of Psychology, Beijing 100101, China

³Department of Psychology, University of Chinese Academy of Sciences, Beijing 100049, China

⁴Department of Cognitive Neuroscience, Donders Institute, Radboud University Medical Centre, Nijmegen
6500 HD, The Netherlands

⁵Magnetic Resonance Imaging Research Center, Institute of Psychology, Chinese Academy of Sciences,
Beijing 100101, China

⁶International Big-Data Center for Depression Research, Chinese Academy of Sciences, Beijing 100101,
China

⁷Department of Psychiatry, Yale University School of Medicine, New Haven, CT, USA

⁸School of Neuroscience and Neuroimaging, Sino-Danish College, University of Chinese Academy of
Sciences, Beijing 101408, China

⁹Sino-Danish Center, University of Chinese Academy of Sciences, 100190, Beijing, China

¹⁰Danish Research Centre for Magnetic Resonance, Centre for Functional and Diagnostic Imaging and
Research, Copenhagen University Hospital Hvidovre, Hvidovre, Denmark

¹¹Department of Applied Mathematics and Computer Science, Technical University of Denmark, Kongens
Lyngby, Denmark

¹²Beijing Key Laboratory of Learning and Cognition, School of Psychology, Capital Normal University,
Beijing 100048, China

¹³Affiliated Brain Hospital of Guangzhou Medical University, Guangzhou, Guangdong 510370, China

¹⁴The First Affiliated Hospital of Jinan University, Guangzhou, Guangdong 250024, China

¹⁵Department of Radiology, The First Affiliated Hospital, College of Medicine, Zhejiang University, Hangzhou, Zhejiang 310058, China

¹⁶Department of Radiology and Huaxi MR Research Center (HMRRC), Functional and Molecular Imaging Key Laboratory of Sichuan Province, West China Hospital of Sichuan University, Chengdu, Sichuan 610041, China

¹⁷Research Unit of Psychoradiology, Chinese Academy of Medical Sciences, Chengdu, Sichuan 610041, China

¹⁸Department of Psychiatry, First Affiliated Hospital of Kunming Medical University, Kunming, Yunnan 650032, China

¹⁹Department of Psychiatry, and National Clinical Research Center for Mental Disorders, The Second Xiangya Hospital of Central South University, Changsha, Hunan 410011, China

²⁰Department of Neurology, Affiliated ZhongDa Hospital of Southeast University, Nanjing, Jiangsu 210009, China

²¹Department of Psychiatry, First Affiliated Hospital of Chongqing Medical University, Chongqing 400042, China

²²Department of Clinical Psychology, Suzhou Psychiatric Hospital, The Affiliated Guangji Hospital of Soochow University, Suzhou, Jiangsu 215003, China

²³Xijing Hospital of Air Force Military Medical University, Xi'an, Shaanxi 710032, China

²⁴Beijing Anding Hospital, Capital Medical University, Beijing 100120, China

²⁵The Third Affiliated Hospital of Zhengzhou University, Zhengzhou, Henan 450052, China

²⁶Affiliated Mental Health Center & Hangzhou Seventh People's Hospital, Zhejiang University School of Medicine, Hangzhou, Zhejiang 310063, China

²⁷Mental Health Center and Psychiatric Laboratory, West China Hospital of Sichuan University, Chengdu, Sichuan 610044, China

²⁸Department of Psychosomatics and Psychiatry, Zhongda Hospital, School of Medicine, Southeast University, Nanjing, Jiangsu 210009, China

²⁹Shenzhen Kangning Hospital, Shenzhen, Guangdong 518020, China

³⁰Faculty of Psychology, Southwest University, Chongqing 400715, China

³¹National Clinical Research Center for Mental Disorders (Peking University Sixth Hospital) & Key Laboratory of Mental Health, Ministry of Health (Peking University), Beijing 100191, China

³²Xi'an Central Hospital, Xi'an, Shaanxi 710004, China

³³Institute of Neuroscience, Chongqing Medical University, Chongqing 400016, China

³⁴Chongqing Key Laboratory of Neurobiology, Chongqing 400000, China

³⁵Department of Neurology, First Affiliated Hospital of Chongqing Medical University, Chongqing 400042, China

³⁶Department of Radiology, First Affiliated Hospital of Anhui Medical University, Hefei, Anhui 230022, China

³⁷Research Center of Clinical Medical Imaging, Hefei, Anhui 230032, China

³⁸Anhui Provincial Institute of Translational Medicine, Hefei, Anhui 230032, China

³⁹West China Hospital of Sichuan University, Chengdu, Sichuan 610044, China

⁴⁰First Hospital of Shanxi Medical University, Taiyuan, Shanxi 030001, China

⁴¹Developmental Population Neuroscience Research Center, IDG/McGovern Institute for Brain Research, Beijing Normal University, Beijing 100091, China

⁴²National Basic Science Data Center, Beijing 100038, China

*A list of authors and their affiliations appears at the end of the paper.

#Corresponding authors:

Xiao-Ping Wu, M.D.

Xi'an Central Hospital, Xi'an, Shaanxi 710004, China

E-mail: szping518@163.com

or

Chao-Gan Yan, Ph.D.

Department of Psychological and Cognitive Sciences, Tsinghua University, Beijing, 100084, China

E-mail: yancg@tsinghua.edu.cn

Running Title: Dynamic Functional Architecture in Major Depressive Disorder

Abstract

Major depressive disorder (MDD) imposes significant global health burdens, yet its underlying neural mechanisms remain elusive. Traditional static functional metrics inadequately capture the brain's dynamic nature, motivating the exploration of dynamic functional metrics to understand both the temporal and spatial reconfigurations of brain networks in MDD. Leveraging the Depression Imaging Research Consortium (DIRECT) dataset, this study conducted vertex-wise dynamic analyses in a large cohort of MDD patients ($n = 1\,660$) and healthy controls ($n = 1\,341$). We identified significant alterations in temporal stability across the brain, with MDD patients exhibiting increased stability in higher-order association areas (e.g., frontoparietal and default mode networks) and decreased stability in primary sensory-motor regions. Among the regions showing altered temporal stability, brain-symptom relationships were further explored. We identified a set of brain regions including the superior frontal gyrus, postcentral gyrus and superior insular sulcus, which were potentially involved in the common abnormal dFC network and associated with insomnia, feelings of guilt, and insight symptoms in MDD. By incorporating advanced vertex-wise dynamic functional analyses and a large sample size, this study provides insights into the neural mechanisms of MDD, emphasizing the value of dynamic approaches for identifying biomarkers. Future longitudinal and task-

based studies are promising to elucidate causal relationships and refine personalized therapeutic interventions targeting specific dynamic dysfunctions in MDD.

ARTICLE IN PRESS

Introduction

Major depressive disorder (MDD) has emerged as a prevalent mental health issue worldwide for decades. The global incidence of depression has increased by approximately 50%, with no reduction in its prevalence or burden since it affected 172 million individuals in 1990 ^[1, 2]. MDD accounts for 49.4 million disability-adjusted life-years (DALYs) in a single year ^[3], imposing significant burdens and suffering on individual patients while also profoundly impacting their families and society as a whole.

Despite extensive research efforts, the underlying causes of depression and the associated changes in brain function still remain elusive ^[4]. Increasing evidence suggests that depression could be viewed as involving multiple symptom dimensions or clusters, a perspective often referred to as multidimensionality ^[5-7]. These different symptom clusters appear to vary in their responsiveness to treatments ^[6, 8]. For instance, dysphoric symptoms and anxiety symptoms respond best to stimulation of distinct brain circuits through transcranial magnetic stimulation (TMS) in the treatment of depression ^[6, 9]. Depression is not a single, uniform disorder, but rather a constellation of symptoms that may require targeted, symptom-specific treatments based on the unique neurobiological mechanisms underlying different symptom clusters ^[10].

MDD may involve multiple dysfunctions in emotional, cognitive, and somatic domains, engaging

various brain regions, and this complexity has been accompanied by diverse methodological and quantitative approaches in the research literature ^[11-15]. In a recent systematic review ^[13], we documented previous fMRI studies on MDD that avoided several common methodological pitfalls, in which almost all studies focused on static metrics. For instance, static functional connectivity has been primarily used as a simple yet effective indicator for evaluating functional brain networks, which are generally assumed to be relatively stable over the observed timescales, although not necessarily corresponding to direct structural connectivity ^[16-21]. Despite their utility, such static measures may overlook the brain's dynamic nature. Neural activity fluctuates continuously, even during so-called "resting states" ^[22]. Clearly, functional metrics depends on how it is measured, and metrics measured over shorter timescales may be more informative about temporal fluctuations of the brain ^[18]. In other word, dynamic metrics have the benefit of characterizing both the spatial and temporal fluctuations of functional brain signals ^[23-25].

Several metrics have emerged to characterize the dynamic functional patterns ^[24, 26]. DFC variance or standard deviation (SD) refers to the time-varying changes in the strength of specific functional connectivity between brain regions, which is often used to measure the variability in the strength of dFCs ^[27, 28]. However, some studies have reported that the reliability of dFC variance or SD may be relatively limited ^[29, 30]. On the

other hand, to characterize the consistency of these dFCs, temporal stability provides a measure of how stable the brain's functional architecture is during a given period ^[26]. Dynamic functional architecture is critical for the brain's ability to sustain consistent information processing and integration across time, which is essential for maintaining conscious states ^[31, 32]. Previous works ^[26, 28] have mapped the whole-brain distribution pattern of functional stability in healthy individuals and thoroughly confirmed its reproducibility across different computational parameters and scanning conditions. These works found that healthy individuals exhibit high levels of temporal stability in associative brain regions, including the dorsolateral prefrontal cortex, angular gyrus, anterior insula, and superior occipital gyrus ^[26, 28]. In addition, a study proposed an intermediate (meso) scale analysis that characterized temporal variability in a different functional architecture across mental disorders and healthy participants, showing close relationships between variability changes and symptom scores ^[33]. However, the whole-brain distribution pattern of functional stability in patients with depression remains to be characterized.

In MDD, the role of these dynamic measures has gained increasing attention. Previous findings suggested that depression could be associated with a global-level decreased temporal stability, in which sensory perception regions, default mode regions and subcortical regions are most prominently involved

[34-38]. In adolescents with varying severities of depressive symptoms, abnormal frontoinsula dynamics, mainly characterized by increased dominance and transition frequency of frontoinsula-default network states, were associated with higher depression and trait rumination [39], a process that engages dynamic cognitive functions such as self-reflection and monitoring of one's internal state [40]. In addition, the relatively contradictory finding exists. In a study that quantified the dispersion of dFCs in MDD, researchers found that MDD patients showed significantly decreased dFC variability between the frontoparietal and default mode networks [41]. The existing studies suggest that abnormalities in dynamic functional stability and connectivity patterns may underlie the neurobiological mechanisms of dysfunctions in depression, especially in processing self-referential and internally directed information [42, 43]. However, studies in this field remain relatively limited and inconsistent. Previous studies were predominantly conducted using volume-based fMRI methods, which have been shown to degrade the spatial precision of cortical parcellation and the alignment of functional areas compared with surface-based approaches. Due to the complex topology of the cortical surface as a highly folded 2-D sheet, vertex-wise fMRI methods in surface space provide a principled approach, offering spatial localization nearly three times better than traditional voxel-wise methods [44]. Larger-scale, vertex-wise studies are needed to further understand dynamic brain

activities in MDD.

Recently, the Depression Imaging REsearch ConsorTium (DIRECT) ^[45] was established to investigate the brain alterations associated with MDD. The consortium has initially focused on large-scale cohorts of Chinese participants with MDD and healthy controls. Additionally, a state-of-the-art surface-based preprocessing pipeline was introduced in its data sharing process, enhancing the sensitivity of fMRI analysis and enabling more accurate assessment of intrinsic brain functional characteristics in resting state ^[44, 46].

In this study, leveraging the large dataset provided by DIRECT, we employed a cross-sectional design to examine the dynamic alterations in brain functional connectivity between MDD patients and healthy individuals, and further explore the relationship between dynamic functional architecture and clinical symptoms. Using precise vertex-wise measures of dynamic methods, we hypothesize the following: (1) significant differences in the dynamic functional architecture of the brain could exist between MDD patients and healthy controls, and (2) the altered dynamic functional networks could be potentially associated with specific clinical features in MDD, especially those likely involving self-referential and internally-directed cognitive functions such as self-awareness, self-consciousness or self-regulation.

Methods

Participants

This study employed data from the Depression Imaging REsearch ConsorTium (DIRECT) ^[45, 47], consisting of 1 660 MDD patients and 1 341 healthy controls. Detailed information on the site distribution and related data acquisition protocols can be found in Chen et al., 2023 ^[45, 47]. All participants underwent both structural and resting-state functional MRI scans. Depression severity in patients was evaluated by clinicians using the Hamilton Depression Rating Scale (HAMD). Along with the preprocessed MRI data, additional demographic information (including age, sex, and education) and clinical details (such as whether it was the first episode or recurrent, medication use, and illness severity) were provided for analysis.

Imaging quality control

To ensure the quality of neuroimaging data, we considered several factors including head motion index, surface reconstruction quality, spatial registration quality, and brain coverage. Participants' fMRI data were excluded if their mean frame-wise displacement (FD) exceeded 0.2 mm ^[48], if the surface reconstruction or registration was low-quality, or if brain coverage was insufficient. Additionally, to meet the requirements of subsequent analyses, we ensured that the data included complete demographic and site information.

Ultimately, data from a total of 2 891 participants were analyzed, consisting of 1 583 individuals with MDD and 1 308 healthy controls (HC).

MRI data preprocessing

MRI data were preprocessed through the DIRECT data-sharing protocol in accordance with the standardized DPABISurf pipeline ^[45, 49]. Initially, for the functional images, the first ten time points of the functional data were discarded to account for signal stabilization. The fMRIPrep tool ^[46] was then used for the realignment and slice-timing correction, and further steps involved spatial normalization, brain tissue segmentation, and surface reconstruction of the cortical surface. Spatial registration between functional and structural images was then performed, with the functional data being projected into FreeSurfer's fsaverage5 surface space ^[50, 51] (using the standard fsaverage5 surface template containing 10 242 vertices per hemisphere) for cortical surface-based analyses and MNI space for subcortical regions. Nuisance regression for cerebrospinal fluid and white matter signals, along with band-pass filtering (0.01 - 0.1 Hz), were conducted through DPABISurf to remove noises, followed by spatial smoothing. Linear drift was also removed before calculating dynamic metrics.

Stability of dynamic functional architecture

To assess the dynamic functional architecture of the whole-brain functional network in patients with MDD and healthy controls, temporal stability was initially computed align with the previous work ^[26] using the DPABISurf toolbox ^[49] (available with DPABI V7.0 at <http://www.rfmri.org/dpabi>) in surface space (see Figure 1). A sliding window approach ^[25, 52] (window length = 30 TRs, step size = 1 TR) was employed to calculate the time-varying functional connectivity between each vertex across the entire brain during continuous time intervals. This produced a series of dynamic functional connectivity maps for each vertex within each time window. Subsequently, the dynamic functional connectivity strength, represented by Pearson correlation values (transformed using Fisher's r-to-z), was ranked across all time windows, and Kendall's coefficient of concordance (W) was calculated for each vertex. This coefficient, reflecting temporal consistency, was then z-standardized ^[53] to yield a stability index (for the distribution of indices, see Supplementary Figure S4). Higher stability indicates greater temporal consistency in the dynamic network topology of that region.

Initially, one-sample t-tests were performed separately on the Z-standardized stability values in both the healthy control (HC) and major depressive disorder (MDD) groups to characterize the cortical functional organization in both populations. Subsequently, two-sample t-test were conducted to compare functional

stability between the MDD vs. HC, controlling for age, sex, education, head motion (mean FD), and scan site as covariates. Multiple comparison correction was applied using 5 000 permutation tests (PT) with threshold-free cluster enhancement (TFCE) to balance false positive rates and reproducibility of results ^[54].

The cluster-wise threshold was set at $P < 0.025$.

Identification of symptom-associated regions of interest

This is an exploratory step to identify potential regions of interest with altered temporal stability that may participate in the shared abnormal dFC network contributing to the observed differences in stability alterations. We assume two complementary forms of brain-symptom relationships: (i) symptom-related distributed regions, referring to cases where stability values of distributed brain regions are associated with the same symptom (where insomnia is regarded as a unified symptom representation), suggesting that these regions may be more likely to be involve in common abnormal dFC networks; and (ii) symptom-overlapping regions, referring to cases where different symptoms are associated with or “overlap in” the same brain region, such that all regions associated with these overlapping symptoms may also potentially reflect shared abnormal dFC networks. We first extracted the peak values and locations of temporal stability in MDD from the significant clusters obtained in the above two-sample t-test. Partial Spearman correlations

were then conducted to explore the relationship between the stability values and the clinical manifestations of MDD, controlling for age, sex, education years, head motion, and scan site as covariates. We focused on two kinds of clinical features of the disorder: the overall severity of depression, as indicated by the total score on the HAM-D-17 scale, and the severity of specific depressive symptoms, assessed by the individual item scores of the HAM-D-17. Locations were considered potentially sensitive to or associated with a specific clinical feature when the partial correlation coefficient was statistically significant (uncorrected $p < 0.05$). Regions sensitive to the same symptom or the overlapping symptoms were identified as a set of symptom-associated regions of interest, which potentially reflect the shared aberrant dFC network in MDD. Thus, these regions were subsequently used as seed regions for further analyses on dynamic dysfunctional networks.

Seed-based dynamic functional connectivity networks

We next performed exploratory seed-based dFC analyses using the symptom-associated regions of interest. The seed regions were defined by the peak locations of significant clusters obtained from the comparison of temporal stability in MDD vs. HC, and were further identified by the above exploratory correlation analyses as potential components of specific symptom-brain relationships. FCs between the

seed location and all other vertices across the whole brain within each time window were generated with the DPABISurf toolbox ^[49] (available with DPABI V7.0 at <http://www.rfmri.org/dpabi>) using the same method as in the previous temporal stability analyses, maintaining the same window length and step size. The z-standardized variation across all time windows of the dFC maps was then calculated vertex by vertex using the standard deviation (SD), resulting in the seed-based dFC map for each participant. This metric reflects variability in the strength of functional connectivity between brain regions ^[27, 28], and could be used to explore which connections specifically contributed to the differences in the observed stability alterations ^[26]. Two-sample t-tests were performed to identify abnormal dFC distributions in MDD by comparing seed-based dFC maps between MDD vs. HC. As with the previous stability analysis, age, sex, education, head motion, and scan site were controlled as covariates. Multiple comparison correction was applied using 5 000 permutations with TFCE, and the cluster-wise threshold was set at $P < 0.05$.

Complementary analysis

Given the potential impact of window-length settings on the results of the sliding-window approach, we further repeated the stability analysis with two window length of 30 s and 120 s (15 TRs and 60 TRs) to test the robustness of our findings. Additionally, we assessed whether the identified components of specific

brain-symptom relationships were likely to participant in the shared abnormal dFC network. We used an additional seed located within both the potentially shared network and the significant stability clusters to independently generate the altered dFCs between MDD and HC, and to test whether this abnormal pattern involved the identified set of symptom-associated brain regions and resembled the previously observed altered connections. (For details, see Supplementary Material.)

Results

Profile of functional stability across participants

To characterize the general stability in HC and MDD patients, we first conducted one-sample t-tests against 0 on the z-standardized stability metrics. We found that functional stability was higher in the bilateral superior and middle frontal cortex, anterior part of the superior and central insular gyri and sulci, angular gyrus, posterior cingulate cortex, precuneus, parieto-occipital cortex, superior occipital cortex, cuneus, lingual gyrus, lateral temporal cortex, and lower in central sulcus, pre- and postcentral gyri, inferior insular sulcus, anterior temporal lobe, orbitofrontal cortex, parahippocampal gyrus, and anterior and middle

cingulate cortex. Similar dynamic functional profiles were revealed in both HC (see Figure 2A) and MDD (see Figure 2B).

Altered temporal stability in MDD

Compared to healthy subjects, MDD patients exhibited decreased stability in brain areas of bilateral precentral gyri, central sulci, postcentral gyri, and sulci, superior parietal gyri, superior insular sulci, anterior, inferior, middle and superior occipital gyri and sulci, parieto-occipital fissures, cuneus, calcarine fissures, lingual gyri and sulci, and right paracentral lobule, which primarily belongs to somatomotor network (SN) and visual network (VN) ^[55]; and increased stability in brain areas of bilateral superior and middle frontal sulci and gyri, inferior insular sulci, inferior temporal sulci and gyri, orbitofrontal sulci and gyri, anterior cingulate gyri, rectus gyri, temporal poles, and parahippocampal gyri, right middle temporal gyrus and temporal transverse gyrus, and left inferior frontal sulcus, occipital pole and precuneus, which involved most frontoparietal network (FPN) and limbic network (LN), along with several regions of default mode network (DMN). See Figure 3 and Supplementary Material, Table S1.

Identification of symptom-associated regions of interest

We further performed exploratory analyses on the correlation between patients' HAMD scores and their functional stability to identify key brain regions that have the potential to be sensitive to clinical features and indicate the specific underlying dysfunctional networks. The the stability values and locations were derived from the peaks of each significant cluster in the previous two-sample t-test comparing stability between MDD vs. HC, including bilateral postcentral gyri, superior insula sulci, superior frontal gyri, left occipital pole, superior occipital gyrus, right inferior temporal gyrus, temporal transverse gyrus, middle occipital gyrus, and superior frontal sulcus (see Figure 4A). A total of 1 314 MDD patients with HAMD total scores and 968 patients with item scores were included in the Spearman partial correlation analysis. We observed no significant correlation between the total HAMD score of patients with either of these seeds. As for the severity of specific symptoms, HAMD scores of feelings of guilt in MDD patients significantly correlated with their stability values in left superior frontal gyrus ($r=0.010$, $p=0.004$) and in right superior frontal gyrus ($r=-0.08$, $p=0.012$). Meanwhile, HAMD scores of early insomnia also negatively correlated with stability in right superior frontal gyrus ($r=-0.09$, $p=0.008$). And HAMD scores of middle insomnia negatively correlated with stability in right postcentral gyrus ($r=-0.07$, $p=0.045$) and in right superior insular cortex ($r=-0.08$, $p=0.012$), along with negative correlation between HAMD scores of insight and stability in this region

($r=-0.08$, $p=0.011$). Moreover, HAMD scores of gastrointestinal somatic symptoms significantly correlated with stability values extracted from left superior insular cortex ($r=-0.08$, $p=0.012$). And significant correlations were observed between scores of general somatic symptoms and stability in right superior frontal sulcus ($r=0.07$, $p=0.026$), and between scores of depressed mood and right middle occipital gyrus ($r=-0.09$, $p=0.010$). See Figure 4B and 4C.

Among these results, we observed that distributed brain regions could be sensitive to the same symptom, such as the relationship between insomnia and right superior insula, right postcentral gyrus and right superior frontal gyrus; meanwhile different symptoms could also overlap within the same brain region, such as the brain-symptom relationship between feelings of guilt, insomnia and right superior frontal gyrus, suggesting these brain regions and symptoms may link in a meaningful way. It should be noted that the current analysis alone is not sufficient to determine whether the identified brain-symptom relationships reflect correlation or prediction - that is, whether symptom scores can predict stability values, or vice versa. Thus, we identified a set of brain regions including bilateral superior frontal gyri, right postcentral gyrus and right superior insular sulcus which potentially shared the abnormal dFC network associated with insomnia, feelings of guilt, and insight symptoms in MDD.

Dynamic dysfunctional connectivity networks in MDD

Peak locations derived from the significant stability clusters and further identified as symptom-associated regions of interest were sequentially used as seeds to probe the specific dFC variability alterations between MDD and HC that may primarily contribute to the stability differences, and to explore their potentially shared abnormal dFC networks. Compared to healthy controls, MDD patients exhibited significantly decreased variability in dynamic functional connectivity between left postcentral gyrus and bilateral superior frontal gyri (see Figure 5A and 5B, and Supplementary Material, Table S2 and Table S3). Meanwhile, significantly increased dynamic functional connectivity variation was observed between right superior frontal gyrus and sulcus, right middle frontal gyrus and sulcus, as well as right precentral gyrus (see Figure 5B and Supplementary Material, Table S3). Additionally, MDD patients showed decreased dynamic connectivity variability between right postcentral gyrus and bilateral anterior and middle cingulate gyri and sulci, along with significantly increase within bilateral central sulci, postcentral gyri and right precentral gyrus (see Figure 5C and Supplementary Material, Table S4). And we observed significantly increased dynamic functional connectivity variability between right superior insula with right superior frontal gyrus, and middle cingulate gyrus and sulcus (see Figure 5D and Supplementary Material, Table S5).

Complementary analysis

In complementarity, we tested two alternative window lengths (15 TRs and 60 TRs) by repeating the dynamic stability analysis of the current study. The findings largely replicated our primary analysis (see Supplementary material, Figure S1 and Figure S2), supporting the robustness of our main results. The effect tended to be slightly less prominent when using a 15-TR window length (Figure S1) and more prominent when using a 60-TR window length (Figure S2). A previous study suggested that the test-retest reliability of dFC analyses may follow a monotonic increasing and then decreasing function of window length, with an early peak observed at approximately 20 to 50 TRs^[30]. Thus, we presented our main results using a 30-TR (60 s) sliding window to achieve potentially optimal performance.

Additionally, to assess whether the four identified symptom-associated brain regions are likely to reflect the shared abnormal dFC network, the left PCG ($x=-57$, $y=-16$, $z=42$), as a potential key node of the network, was used as an additional seed to replicate the seed-based dFC analysis results. Compared to healthy controls, significant alterations of dFC variability in MDD patients involved the bilateral postcentral gyrus, superior and middle frontal cortex, insula, anterior and middle cingulate cortex, as well as the right precentral gyrus (see Supplementary Material, Figure S3A). These significant clusters overlapped with the

altered dFCs observed based on the four symptom-associated brain regions and approximately covered all of them (see Supplementary Material, Figure S3B).

Discussion

The present study provides a comprehensive characterization of the altered dynamic functional architecture in MDD using vertex-wise dynamic fMRI analyses in a large sample size. We first depicted consistent dynamic functional profiles across participants and further revealed increased stability in brain regions of FPN, LN and DMN, alongside decreased stability in regions of SMN and VN in MDD compared to HC. Then we highlighted the potential brain-symptom relationships and identified a set of symptom-associated brain regions, including bilateral superior frontal gyri, the right postcentral gyrus, and the right superior insular sulcus, which were sensitive to insomnia, feelings of guilt, and insight symptoms in MDD. Further examinations of the distributed patterns of the altered dFCs revealed notable overlaps, suggesting the potentially common aberrant dynamic functional network associated with specific clinical features of MDD, which involves the post- and precentral gyri, central sulcus, superior and middle frontal cortex, anterior and middle cingulate cortex, and insular cortex.

From a dynamic perspective, higher-order association hubs of the brain may play a role in consistent temporal coordination with other regions to maintain stable functional organization, which may be associated with their within-state increased stability ^[26, 56]. Conversely, primary sensory and motor regions are primarily engaged in responding to rapidly changing stimuli from the external environment, leading to a more flexible and transient dynamic profile ^[17, 56, 57]. In the present study, the stability profiles obtained through one-sample t-tests align with previous findings in healthy participants ^[26, 28], showing higher stability in frontal cortex, posterior cingulate cortex, occipital cortex, and anterior insula, which are likely involved in stable, higher-order cognitive functions, and relatively lower stability in somatosensory cortex, orbitofrontal cortex, and transverse temporal cortex, associated with more flexible, context-dependent processing. This consistency with previous findings supports the reproducibility and reliability of our results. Furthermore, the similar stability profiles observed across both healthy and depressive groups suggest a shared dynamic functional organization trend irrespective of disease status.

While compared to healthy participants, MDD patients showed significantly decreased stability in widespread regions of the primary sensory and motor cortex, as well as a few association areas, such as the superior insular regions. Conversely, they exhibited increased stability in higher-order functional and

association areas, as well as parts of the primary auditory cortex. This pattern of widespread decreased stability in primary functional areas and increased stability in higher-order regions may suggest an unbalanced trade-off between externally and internally directed cognition when processing external and internal information in MDD patients.

Externally directed cognition focuses on processing external environmental stimuli, while internally directed cognition focus more on self-referential thoughts, memories, and mental imagery ^[58-60]. The dynamic feature in BOLD signals across widespread cortical areas, including regions of the VAN, DMN, DAN, and FPN, is thought to reflect internally and externally directed cognitive processes ^[58, 61-63]. Previous studies have implicated these brain regions and networks in MDD and other mood disorders, yet the precise dynamic properties of these regions remain largely unexplored ^[39, 64-67]. Notably, existing evidence from healthy participants align with this view. In studies included younger and older adults, reduced modulation of dynamic variability across brain regions and conditions was associated with poorer cognitive performance and older age, indicating a reduction in the flexibility of dynamic variability associated with less ability to switch between internal and external brain states ^[61]. Additionally, the increased stability in higher-order cognitive and association areas could be linked to a preference for internally directed cognition,

as studies in healthy individuals have shown that reduced BOLD variability is associated with better performance in internally-directed tasks ^[68].

Furthermore, the common distribution pattern observed in the altered dFCs derived from the exploratory symptom-associated brain regions suggests a potentially meaningful cluster of specific brain-symptom relationships. Consistent with previous discussions, impaired processing of external and internal information, or disrupted cognition and regulation of one's own states, may help explain depressive symptoms such as feelings of guilt and insight-related symptoms. Due to the retrospective nature of the resting-state fMRI dataset, task-based data for directly explaining the dysfunctional mechanisms were not included in this study. However, existing task-based studies can support our findings. Specifically, the insular cortex, as a higher-order association hub, is widely implicated in subjective bodily sensations and plays a prominent role in processing a wide range of emotional experiences, especially self-generated emotions ^[67, 69-71]. For instance, a study on tactile perception in healthy individuals shows that the insular cortex is activated only during self-perceived touch but not when observing others being touched, underscoring its role in subjective awareness ^[72]. Moreover, feelings of guilt in depression is often associated with trait rumination, which characterized by an elevated repetitive, self-reflective, and passive

focus on the negative status ^[40, 73]. Significant changes were described in the degree centrality of the left superior frontal gyrus and in the functional connectivity between the superior frontal gyrus, the frontoparietal network, and the dorsal attention network during the in-scanner rumination task in MDD ^[74]. It is thought that powerful and negative affective reaction to internal cues could result in a funneling effect and substantially reduce the breadth of the associative network ^[75].

Additionally, insomnia emerged as an overlapping symptom linked to altered dFCs in the superior insula, superior frontal cortex, and postcentral cortex, suggesting an potentially underlying mechanism of failed transitions between internally and externally directed cognition. This aligns with previous evidence of altered activations in related brain regions in depressed patients, including the prefrontal, sensorimotor, and insular cortices during REM sleep, which may be related to cognitive and affective dysregulation ^[76, 77].

Neurochemical evidence showed alterations in norepinephrine levels, which are associated with arousal-related emotional processes ^[78, 79]. Furthermore, studies on static functional connectivity reported alterations across widespread networks in MDD associated with poor sleep quality ^[80-83]. REM sleep is a critical stage for the whole-night sleep, involving multidimensional functions such as emotion regulation, recognition, reasoning, and memory, which requires the coordinated activity of extensive brain networks ^{[84-}

^{86]}. Dysregulated functional connectivity in MDD may disrupt these processes, exacerbating the vicious cycle of poor sleep and worsening cognitive and mood symptoms ^[87]. The identification of the abnormal dFC distribution, along with the potential associations with early insomnia and nocturnal wakefulness, shows its exploratory value as the potential predictor and therapeutic target for managing sleep disturbances in MDD ^[88-90].

Our findings highlight specific regions where disrupted temporal reconfiguration of functional networks is associated with a set of depressive symptoms, offering insights into the specific dynamic neural mechanism underlying clinical features of MDD. It is important to note that although we used significant correlations to identify dynamic changes in brain regions that are sensitive to particular clinical features, this alone does not confirm the direct correlation model between specific brain regions and symptoms. Instead, it allows us to identify a set of exploratory components of specific brain-symptom relationships. The convergence of these symptom-associated regions and circuits with key brain areas targeted by Deep Brain Stimulation (DBS) and Transcranial Magnetic Stimulation (TMS) in treatment-resistant depression patients, including changes in the frontal and limbic-paralimbic regions (such as the dorsolateral prefrontal cortex, subgenual cingulate, posterior cingulate cortex, and insula), supports the continued refinement of

depression circuit models in future studies ^[91-93]. This approach may optimize treatment targeting, define biomarkers to guide patient selection, and enhance our understanding of therapeutic mechanisms for symptom remission in depression.

Moreover, the methodological strengths of our study are evident in several aspects. First, the use of whole-brain vertex-wise analyses enabled a more precise mapping of the functional architecture compared to volume-based approaches ^[44]. Second, by leveraging dynamic functional metrics, we provided complementary information about how the temporal organization of brain networks is altered in MDD. Third, the large sample size ($n > 1\,500$ for MDD and $n > 1\,300$ for HC) ensured robust statistical power and generalizability, addressing a major limitation of many neuroimaging studies in MDD, which often suffer from small sample sizes and inconsistent findings ^[54, 94].

Despite these strengths, some limitations should be noted. First, the cross-sectional design prevent us from determining the causal direction of the observed alterations in dynamic dysfunctional features and the clinical features in MDD. Longitudinal studies will be necessary to clarify whether these functional changes represent a neural marker of MDD onset, maintenance, or response to treatments. Second, the reliance on retrospective resting-state fMRI data, while suitable for assessing intrinsic functional architecture, may not

fully capture task-dependent dynamics that are helpful to understanding how MDD patients respond to external stimuli or events. Future studies could combine resting-state and task-based fMRI to provide a more comprehensive picture of dynamic brain function in MDD. Finally, the heterogeneity of depressive symptoms and clinical profiles suggests that future studies could investigate whether the observed dynamic patterns are specific to certain subtypes or dimensions of MDD, potentially paving the way for more personalized interventions.

Conclusion

This work provides a comprehensive investigation into the altered dynamic functional architecture of the brain in MDD, leveraging a large dataset and advanced vertex-wise dynamic analyses. Our findings highlight significant alterations of dFC in a set of symptom-associated regions that are sensitive to clinical features such as guilt, insomnia, and insight symptoms, and point to a shared dysfunctional mechanism underlying specific brain-symptom relationships in MDD. These results offer valuable insights into features of dynamic brain functional connectivity in depression and underscore the potential of dynamic metrics as sensitive tools for identifying specific neural alterations which may be associated with depressive clinical features. Future longitudinal and task-based studies are warranted to refine our understanding of dynamic

functional features in MDD and hold promise for the development of personalized interventions targeting specific dynamic functional networks to alleviate particular clinical symptoms.

Data Availability

The data used in this study are from the DIRECT dataset. The data are publicly available via DIRECT Phase II data-sharing Stage 2 from January 1, 2026 (for access, see <https://rfmri.org/DIRECTPhase2>).

Acknowledgements

This work was supported by Science and Technology Program of Xi'an grant (No. 23YXYJ0004), Xi'an Elite Innovative Talents Program (No. XAYC240059), the Sci-Tech Innovation 2030 - Major Project of Brain Science and Brain-inspired Intelligence Technology (No. 2021ZD0200600), the National Natural Science Foundation of China (No. 82122035, and No. 32300933), Beijing Nova Program of Science and Technology (No. 20230484465), and Beijing Natural Science Foundation (No. J230040).

Ethics Declarations

Competing interests

The authors declare that they have no conflicts of interest to disclose.

Ethics approval

The research protocols were approved by the local ethics review committees of all participating sites, and all participants provided written informed consent before the DIRECT Phase II data sharing. The analysis proposal and the coordinated dataset used in the current study have been reviewed and approved by DIRECT consortium in its data-sharing Stage 1. At this stage, the consortium ensures that proposals have no conflicts before approval, after which this proposal enters the pool of approved proposals to prevent future conflict. The DIRECT Phase II data-sharing Stage 2 will involve unrestricted sharing after January 1st, 2026.

Author Contributions

The authors' responsibilities were as follows – CGY, XPW and XYL conceived, designed, and approved the research plan; all authors contributed to data acquisition and data entry into the dataset; BL, CGY, XYL,

ZHW, YWW, YFL, ZJYH, CNW, HLW, QLG, YRC, SCC, XYW, HLL and ZDL: organized and preprocessed the data; XYL performed the data analyses; XYL, CGY, XC and BL: interpreted the data; XYL wrote the manuscript; XYL, CGY and XPW revised the manuscript; XPW and CGY had primary responsibility for the final content; and all authors read and approved the final manuscript.

DIRECT consortium

Chao-Gan Yan, Ph.D.^{1,2,3,5,6,8,9}, Xiao Chen, Ph.D.^{2,3,5,4}, Li-Ping Cao, M.D.¹³, Wei Chen, M.D.⁴³, Yu-Qi Cheng, M.D.¹⁸, Yi-Ru Fang, M.D.⁴⁴, Qi-Yong Gong, M.D.^{16,17}, Wen-Bin Guo, M.D.¹⁹, Li Kuang, M.D.²¹, Bao-Juan Li, M.D.²³, Tao Li, M.D.^{26,27}, Yan-Song Liu, M.D.²², Zhe-Ning Liu, M.D.¹⁹, Jian-Ping Lu, M.D.²⁹, Qing-Hua Luo, M.D.²¹, Hua-Qing Meng, M.D.²¹, Dai-Hui Peng, M.D.⁴⁴, Jiang Qiu, Ph.D.³⁰, Yue-Di Shen, M.D.⁴⁵, Tian-Mei Si, M.D.³¹, Yan-Qing Tang, M.D.⁴⁶, Chuan-Yue Wang, M.D.²⁴, Fei Wang, M.D.^{46,47}, Hua-Ning Wang, M.D.²³, Kai Wang, M.D.⁵⁰, Xiang Wang, M.D.¹⁹, Ying Wang, M.D.¹⁴, Xiao-Ping Wu, M.D.³², Chun-Ming Xie, M.D.²⁰, Guang-Rong Xie, M.D.¹⁹, Peng Xie, M.D.^{33,34,35}, Xiu-Feng Xu, M.D.¹⁸, Hong Yang, M.D.¹⁵, Jian Yang, M.D.³⁴, Shu-Qiao Yao, M.D.¹⁹, Yong-Qiang Yu, M.D.^{36,37,38}, Yong-Gui Yuan, M.D.²⁸, Ke-Rang Zhang, M.D.⁴⁰, Wei

Zhang, M.D.³⁹, Zhi-Jun Zhang, M.D.²⁰, Jun-Juan Zhu, M.D.⁴⁹, Xi-Nian Zuo, Ph.D.^{41,42}, Jing-Ping Zhao, M.D.¹⁹, Yu-Feng Zang, M.D.^{50,51}

⁴³Department of Psychiatry, Sir Run Run Shaw Hospital, Zhejiang University School of Medicine, Hangzhou, Zhejiang 310020, China

⁴⁴Shanghai Mental Health Center, Shanghai Jiao Tong University School of Medicine, Shanghai 200030, China

⁴⁵Department of Diagnostics, Affiliated Hospital, Hangzhou Normal University Medical School, Hangzhou, Zhejiang 311121, China

⁴⁶Department of Psychiatry, First Affiliated Hospital, China Medical University, Shenyang, Liaoning 110122, China

⁴⁷Early Intervention Unit, Department of Psychiatry, Affiliated Nanjing Brain Hospital, Nanjing Medical University, Nanjing, Jiangsu 210024, China

⁴⁸Department of Neurology, First Affiliated Hospital of Anhui Medical University, Hefei, Anhui 230022, China

⁴⁹Department of Psychiatry, Shanghai Jiao Tong University School of Medicine, Shanghai 200025, China

⁵⁰Center for Cognition and Brain Disorders, Affiliated Hospital of Hangzhou Normal University, Hangzhou, Zhejiang 310018, China

⁵¹Zhejiang Key Laboratory for Research in Assessment of Cognitive Impairments, Hangzhou Zhejiang 310000, China

References

1. Collaborators GDal. **Global burden of 369 diseases and injuries in 204 countries and territories, 1990-2019: a systematic analysis for the Global Burden of Disease Study 2019.** *Lancet* 2020; 396(10258):1204–1222.
2. Liu Q, He H, Yang J, Feng X, Zhao F, Lyu J. **Changes in the global burden of depression from 1990 to 2017: Findings from the Global Burden of Disease study.** *J Psychiatr Res* 2020; 126:134–140.
3. Collaborators C-MD. **Global prevalence and burden of depressive and anxiety disorders in 204 countries and territories in 2020 due to the COVID-19 pandemic.** *Lancet* 2021; 398(10312):1700–1712.
4. Müller VI, Cieslik EC, Serbanescu I, Laird AR, Fox PT, Eickhoff SB. **Altered Brain Activity in Unipolar Depression Revisited: Meta-analyses of Neuroimaging Studies.** *JAMA Psychiatry* 2017; 74(1):47–55.
5. Bagby RM, Ryder AG, Schuller DR, Marshall MB. **The Hamilton Depression Rating Scale: has the gold standard become a lead weight?** *Am J Psychiatry* 2004; 161(12):2163–2177.
6. Kaster TS, Downar J, Vila-Rodriguez F, Baribeau DA, Thorpe KE, Daskalakis ZJ, et al. **Differential symptom cluster responses to repetitive transcranial magnetic stimulation treatment in depression.** *EClinicalMedicine* 2023; 55:101765.
7. Quilty LC, Robinson JJ, Rolland JP, Fruyt FD, Rouillon F, Bagby RM. **The structure of the Montgomery-Åsberg depression rating scale over the course of treatment for depression.** *Int J Methods Psychiatr Res* 2013; 22(3):175–184.
8. Bondar J, Caye A, Chekroud AM, Kieling C. **Symptom clusters in adolescent depression and differential response to treatment: a secondary analysis of the Treatment for Adolescents with Depression Study randomised trial.** *Lancet Psychiatry* 2020; 7(4):337–343.
9. Siddiqi SH, Taylor SF, Cooke D, Pascual-Leone A, George MS, Fox MD. **Distinct Symptom-Specific Treatment Targets for Circuit-Based Neuromodulation.** *Am J Psychiatry* 2020; 177(5):435–446.
10. Ahmed AT, Frye MA, Rush AJ, Biernacka JM, Craighead WE, McDonald WM, et al. **Mapping depression rating scale phenotypes onto research domain criteria (RDoC) to inform biological research in mood disorders.** *J Affect Disord* 2018; 238:1–7.
11. Genon S, Reid A, Langner R, Amunts K, Eickhoff SB. **How to Characterize the Function of a Brain Region.** *Trends Cogn Sci* 2018; 22(4):350–364.
12. Gray JP, Muller VI, Eickhoff SB, Fox PT. **Multimodal Abnormalities of Brain Structure and Function**

in Major Depressive Disorder: A Meta-Analysis of Neuroimaging Studies. *Am J Psychiatry* 2020; 177(5):422–434.

13. Li XY, Chen X, Yan CG. **Altered cerebral activities and functional connectivity in depression: a systematic review of fMRI studies.** *Quantitative Biology* 2022; 10(4):366–380.

14. Assem M, Glasser MF, Van Essen DC, Duncan J. **A Domain-General Cognitive Core Defined in Multimodally Parcellated Human Cortex.** *Cereb Cortex* 2020; 30(8):4361–4380.

15. Fehnel SE, Forsyth BH, DiBenedetti DB, Danchenko N, François C, Brevig T. **Patient-centered assessment of cognitive symptoms of depression.** *CNS Spectr* 2016; 21(1):43–52.

16. Patil AU, Ghate S, Madathil D, Tzeng OJL, Huang HW, Huang CM. **Static and dynamic functional connectivity supports the configuration of brain networks associated with creative cognition.** *Sci Rep* 2021; 11(1):165.

17. Cole MW, Bassett DS, Power JD, Braver TS, Petersen SE. **Intrinsic and task-evoked network architectures of the human brain.** *Neuron* 2014; 83(1):238–251.

18. Park HJ, Friston K. **Structural and functional brain networks: from connections to cognition.** *Science* 2013; 342(6158):1238411.

19. Greicius MD, Supekar K, Menon V, Dougherty RF. **Resting-state functional connectivity reflects structural connectivity in the default mode network.** *Cereb Cortex* 2009; 19(1):72–78.

20. Honey CJ, Sporns O, Cammoun L, Gigandet X, Thiran JP, Meuli R, et al. **Predicting human resting-state functional connectivity from structural connectivity.** *Proc Natl Acad Sci U S A* 2009; 106(6):2035–2040.

21. Markov NT, Misery P, Falchier A, Lamy C, Vezoli J, Quilodran R, et al. **Weight consistency specifies regularities of macaque cortical networks.** *Cereb Cortex* 2011; 21(6):1254–1272.

22. Chang C, Glover GH. **Time-frequency dynamics of resting-state brain connectivity measured with fMRI.** *Neuroimage* 2010; 50(1):81–98.

23. Long Y, Liu X, Liu Z. **Temporal Stability of the Dynamic Resting-State Functional Brain Network: Current Measures, Clinical Research Progress, and Future Perspectives.** *Brain Sci* 2023; 13(3).

24. Preti MG, Bolton TA, Van De Ville D. **The dynamic functional connectome: State-of-the-art and perspectives.** *Neuroimage* 2017; 160:41–54.

25. Hutchison RM, Womelsdorf T, Allen EA, Bandettini PA, Calhoun VD, Corbetta M, et al. **Dynamic functional connectivity: promise, issues, and interpretations.** *Neuroimage* 2013; 80:360–378.

-
26. Li L, Lu B, Yan CG. **Stability of dynamic functional architecture differs between brain networks and states.** *Neuroimage* 2020; 216:116230.
27. Lu F, Chen Y, Cui Q, Guo Y, Pang Y, Luo W, et al. **Shared and distinct patterns of dynamic functional connectivity variability of thalamo-cortical circuit in bipolar depression and major depressive disorder.** *Cereb Cortex* 2023; 33(11):6681–6692.
28. Chen X, Yan CG. **Hypostability in the default mode network and hyperstability in the frontoparietal control network of dynamic functional architecture during rumination.** *Neuroimage* 2021; 241:118427.
29. Choe AS, Nebel MB, Barber AD, Cohen JR, Xu Y, Pekar JJ, et al. **Comparing test-retest reliability of dynamic functional connectivity methods.** *Neuroimage* 2017; 158:155–175.
30. Zhang C, Baum SA, Adduru VR, Biswal BB, Michael AM. **Test-retest reliability of dynamic functional connectivity in resting state fMRI.** *Neuroimage* 2018; 183:907–918.
31. Hasson U, Yang E, Vallines I, Heeger DJ, Rubin N. **A hierarchy of temporal receptive windows in human cortex.** *J Neurosci* 2008; 28(10):2539–2550.
32. Dehaene S, Lau H, Kouider S. **What is consciousness, and could machines have it?** *Science* 2017; 358(6362):486–492.
33. Zhang J, Cheng W, Liu Z, Zhang K, Lei X, Yao Y, et al. **Neural, electrophysiological and anatomical basis of brain-network variability and its characteristic changes in mental disorders.** *Brain* 2016; 139(Pt 8):2307–2321.
34. Long Y, Cao H, Yan C, Chen X, Li L, Castellanos FX, et al. **Altered resting-state dynamic functional brain networks in major depressive disorder: Findings from the REST-meta-MDD consortium.** *Neuroimage Clin* 2020; 26:102163.
35. Ouyang X, Long Y, Wu Z, Liu D, Liu Z, Huang X. **Temporal Stability of Dynamic Default Mode Network Connectivity Negatively Correlates with Suicidality in Major Depressive Disorder.** *Brain Sci* 2022; 12(9).
36. Zhao Z, Zhang Y, Chen N, Li Y, Guo H, Guo M, et al. **Altered temporal reachability highlights the role of sensory perception systems in major depressive disorder.** *Prog Neuropsychopharmacol Biol Psychiatry* 2022; 112:110426.
37. Hou Z, Kong Y, He X, Yin Y, Zhang Y, Yuan Y. **Increased temporal variability of striatum region facilitating the early antidepressant response in patients with major depressive disorder.** *Prog*

Neuropsychopharmacol Biol Psychiatry 2018; 85:39–45.

38. Wise T, Marwood L, Perkins AM, Herane-Vives A, Joules R, Lythgoe DJ, et al. **Instability of default mode network connectivity in major depression: a two-sample confirmation study.** *Transl Psychiatry* 2017; 7(4):e1105.
39. Kaiser RH, Kang MS, Lew Y, Van Der Feen J, Aguirre B, Clegg R, et al. **Abnormal frontoinsula-default network dynamics in adolescent depression and rumination: a preliminary resting-state co-activation pattern analysis.** *Neuropsychopharmacology* 2019; 44(9):1604–1612.
40. Larabi DI, Renken RJ, Cabral J, Marsman JC, Aleman A, Ćurčić-Blake B. **Trait self-reflectiveness relates to time-varying dynamics of resting state functional connectivity and underlying structural connectomes: Role of the default mode network.** *Neuroimage* 2020; 219:116896.
41. Demirtaş M, Tornador C, Falcón C, López-Solà M, Hernández-Ribas R, Pujol J, et al. **Dynamic functional connectivity reveals altered variability in functional connectivity among patients with major depressive disorder.** *Hum Brain Mapp* 2016; 37(8):2918–2930.
42. Whitfield-Gabrieli S, Ford JM. **Default mode network activity and connectivity in psychopathology.** *Annu Rev Clin Psychol* 2012; 8:49–76.
43. Tian S, Zhang S, Mo Z, Chattun MR, Wang Q, Wang L, et al. **Antidepressants normalize brain flexibility associated with multi-dimensional symptoms in major depressive patients.** *Prog Neuropsychopharmacol Biol Psychiatry* 2020; 100:109866.
44. Coalson TS, Van Essen DC, Glasser MF. **The impact of traditional neuroimaging methods on the spatial localization of cortical areas.** *Proc Natl Acad Sci U S A* 2018; 115(27):E6356–e6365.
45. Chen X, Lu B, Li HX, Li XY, Wang YW, Castellanos FX, et al. **The DIRECT consortium and the REST-meta-MDD project: towards neuroimaging biomarkers of major depressive disorder.** *Psychoradiology* 2022; 2(1):32–42.
46. Esteban O, Markiewicz CJ, Blair RW, Moodie CA, Isik AI, Erramuzpe A, et al. **fMRIPrep: a robust preprocessing pipeline for functional MRI.** *Nat Methods* 2019; 16(1):111–116.
47. Chen X, Lu B, Wang YW, Li XY, Wang ZH, Li HX, et al. **Functional Connectivity Abnormalities of the Subgenual Anterior Cingulate Cortex: Implications for Transcranial Magnetic Stimulation in Depression.** *bioRxiv* 2023.
48. Jenkinson M, Bannister P, Brady M, Smith S. **Improved optimization for the robust and accurate linear registration and motion correction of brain images.** *Neuroimage* 2002; 17(2):825–841.

-
49. Yan CG, Wang XD, Lu B. **DPABISurf: data processing & analysis for brain imaging on surface.** *Sci Bull (Beijing)* 2021; 66(24):2453–2455.
50. Dale AM, Fischl B, Sereno MI. **Cortical surface-based analysis. I. Segmentation and surface reconstruction.** *Neuroimage* 1999; 9(2):179–194.
51. Fischl B, Sereno MI, Tootell RB, Dale AM. **High-resolution intersubject averaging and a coordinate system for the cortical surface.** *Hum Brain Mapp* 1999; 8(4):272–284.
52. Hutchison RM, Womelsdorf T, Gati JS, Everling S, Menon RS. **Resting-state networks show dynamic functional connectivity in awake humans and anesthetized macaques.** *Hum Brain Mapp* 2013; 34(9):2154–2177.
53. Yan CG, Craddock RC, Zuo XN, Zang YF, Milham MP. **Standardizing the intrinsic brain: towards robust measurement of inter-individual variation in 1000 functional connectomes.** *Neuroimage* 2013; 80:246–262.
54. Chen X, Lu B, Yan CG. **Reproducibility of R-fMRI metrics on the impact of different strategies for multiple comparison correction and sample sizes.** *Hum Brain Mapp* 2018; 39(1):300–318.
55. Yeo BT, Krienen FM, Sepulcre J, Sabuncu MR, Lashkari D, Hollinshead M, et al. **The organization of the human cerebral cortex estimated by intrinsic functional connectivity.** *J Neurophysiol* 2011; 106(3):1125–1165.
56. Lerner Y, Honey CJ, Silbert LJ, Hasson U. **Topographic mapping of a hierarchy of temporal receptive windows using a narrated story.** *J Neurosci* 2011; 31(8):2906–2915.
57. Cole MW, Pathak S, Schneider W. **Identifying the brain's most globally connected regions.** *Neuroimage* 2010; 49(4):3132–3148.
58. Dixon ML, Fox KC, Christoff K. **A framework for understanding the relationship between externally and internally directed cognition.** *Neuropsychologia* 2014; 62:321–330.
59. Chun MM, Golomb JD, Turk-Browne NB. **A taxonomy of external and internal attention.** *Annu Rev Psychol* 2011; 62:73–101.
60. Petersen SE, Posner MI. **The attention system of the human brain: 20 years after.** *Annu Rev Neurosci* 2012; 35:73–89.
61. Grady CL, Garrett DD. **Brain signal variability is modulated as a function of internal and external demand in younger and older adults.** *Neuroimage* 2018; 169:510–523.
62. Buckner RL, Andrews-Hanna JR, Schacter DL. **The brain's default network: anatomy, function, and**

relevance to disease. *Ann N Y Acad Sci* 2008; 1124:1–38.

63. Deco G, Rolls ET, Romo R. **Stochastic dynamics as a principle of brain function.** *Prog Neurobiol* 2009; 88(1):1–16.

64. Gogolla N. **The insular cortex.** *Curr Biol* 2017; 27(12):R580–r586.

65. Stoychev KR. **Neuroimaging Studies in Patients With Mental Disorder and Co-occurring Substance Use Disorder: Summary of Findings.** *Front Psychiatry* 2019; 10:702.

66. Turel O, He Q, Wei L, Bechara A. **The role of the insula in internet gaming disorder.** *Addict Biol* 2021; 26(2):e12894.

67. Nicolas C, Ju A, Wu Y, Eldirdiri H, Delcasso S, Couderc Y, et al. **Linking emotional valence and anxiety in a mouse insula-amygdala circuit.** *Nat Commun* 2023; 14(1):5073.

68. Roberts RP, Grady CL, Addis DR. **Creative, internally-directed cognition is associated with reduced BOLD variability.** *Neuroimage* 2020; 219:116758.

69. Calder AJ, Keane J, Manes F, Antoun N, Young AW. **Impaired recognition and experience of disgust following brain injury.** *Nat Neurosci* 2000; 3(11):1077–1078.

70. Craig AD. **How do you feel--now? The anterior insula and human awareness.** *Nat Rev Neurosci* 2009; 10(1):59–70.

71. Gasquoin PG. **Contributions of the insula to cognition and emotion.** *Neuropsychol Rev* 2014; 24(2):77–87.

72. McCabe C, Rolls ET, Bilderbeck A, McGlone F. **Cognitive influences on the affective representation of touch and the sight of touch in the human brain.** *Soc Cogn Affect Neurosci* 2008; 3(2):97–108.

73. Marchetti I, Koster EHW, Klinger E, Alloy LB. **Spontaneous Thought and Vulnerability to Mood Disorders: The Dark Side of the Wandering Mind.** *Clin Psychol Sci* 2016; 4(5):835–857.

74. Jia FN, Chen X, Du XD, Tang Z, Ma XY, Ning TT, et al. **Aberrant degree centrality profiles during rumination in major depressive disorder.** *Hum Brain Mapp* 2023; 44(17):6245–6257.

75. Clore G, Gasper K. **Feeling is believing: Some affective influences on belief.** In N. Frijda, A. Manstead, & S. Bem (Eds.), *Emotions and beliefs: How do emotions influence beliefs*. 2000.

76. Nofzinger EA. **Functional neuroimaging of sleep.** *Semin Neurol* 2005; 25(1):9–18.

77. Dang-Vu TT, Schabus M, Desseilles M, Sterpenich V, Bonjean M, Maquet P. **Functional neuroimaging insights into the physiology of human sleep.** *Sleep* 2010; 33(12):1589–1603.

78. Park SP. **In vivo microdialysis measures of extracellular norepinephrine in the rat amygdala**

during sleep-wakefulness. *J Korean Med Sci* 2002; 17(3):395–399.

79. Ouyang M, Hellman K, Abel T, Thomas SA. **Adrenergic signaling plays a critical role in the maintenance of waking and in the regulation of REM sleep.** *J Neurophysiol* 2004; 92(4):2071–2082.
80. Cheng W, Rolls ET, Ruan H, Feng J. **Functional Connectivities in the Brain That Mediate the Association Between Depressive Problems and Sleep Quality.** *JAMA Psychiatry* 2018; 75(10):1052–1061.
81. Fang H, Tu S, Sheng J, Shao A. **Depression in sleep disturbance: A review on a bidirectional relationship, mechanisms and treatment.** *J Cell Mol Med* 2019; 23(4):2324–2332.
82. Gong L, Xu R, Yang D, Wang J, Ding X, Zhang B, et al. **Orbitofrontal Cortex Functional Connectivity-Based Classification for Chronic Insomnia Disorder Patients With Depression Symptoms.** *Front Psychiatry* 2022; 13:907978.
83. Tian Y, Vitiello MV, Wang H, Wang Y, Dong D, Xu H, et al. **Risk of insomnia during COVID-19: effects of depression and brain functional connectivity.** *Cereb Cortex* 2023; 33(11):7015–7025.
84. Dolcos F, LaBar KS, Cabeza R. **Remembering one year later: role of the amygdala and the medial temporal lobe memory system in retrieving emotional memories.** *Proc Natl Acad Sci U S A* 2005; 102(7):2626–2631.
85. Asarnow LD, Manber R. **Cognitive Behavioral Therapy for Insomnia in Depression.** *Sleep Med Clin* 2019; 14(2):177–184.
86. Hu Z, Li W, Ye Y, Zhang F, Liu H, Wang C, et al. **Alterations of functional connectivity in young people with depression mediate the relationship between sleep quality and cognitive function.** *J Affect Disord* 2023; 340:160–166.
87. Goldstein AN, Walker MP. **The role of sleep in emotional brain function.** *Annu Rev Clin Psychol* 2014; 10:679–708.
88. Baglioni C, Battagliese G, Feige B, Spiegelhalter K, Nissen C, Voderholzer U, et al. **Insomnia as a predictor of depression: a meta-analytic evaluation of longitudinal epidemiological studies.** *J Affect Disord* 2011; 135(1-3):10–19.
89. Gebara MA, Siripong N, DiNapoli EA, Maree RD, Germain A, Reynolds CF, et al. **Effect of insomnia treatments on depression: A systematic review and meta-analysis.** *Depress Anxiety* 2018; 35(8):717–731.
90. Grandner MA, Fernandez FX. **The translational neuroscience of sleep: A contextual framework.**

Science 2021; 374(6567):568–573.

91. Cole EJ, Phillips AL, Bentzley BS, Stimpson KH, Nejad R, Barmak F, et al. **Stanford Neuromodulation Therapy (SNT): A Double-Blind Randomized Controlled Trial.** *Am J Psychiatry* 2022; 179(2):132–141.
92. Blumberger DM, Vila-Rodriguez F, Thorpe KE, Feffer K, Noda Y, Giacobbe P, et al. **Effectiveness of theta burst versus high-frequency repetitive transcranial magnetic stimulation in patients with depression (THREE-D): a randomised non-inferiority trial.** *Lancet* 2018; 391(10131):1683–1692.
93. Mayberg HS. **Targeted electrode-based modulation of neural circuits for depression.** *J Clin Invest* 2009; 119(4):717–725.
94. Marek S, Tervo-Clemmens B, Calabro FJ, Montez DF, Kay BP, Hatoum AS, et al. **Reproducible brain-wide association studies require thousands of individuals.** *Nature* 2022; 603(7902):654–660.

Figure legends

Figure 1. Schematic of vertex-to-vertex temporal stability calculation. The schematic diagram illustrates how the stability of dynamic functional architecture is calculated in 2D surface space. For each time window, dynamic functional connectivity (dFC) is calculated between a given vertex and all other vertices across the whole brain within the fsaverage5 surface template. Kendall's concordance coefficient (W) is then computed across all windows to quantify the functional stability of each given vertex. Finally, z-standardization is applied to obtain a transformed stability index for subsequent statistical analyses.

Figure 2. Profile of functional stability across participants. Brain maps of t-statistics illustrate the results of one-sample t-tests on z-standardized stability for **(A)** HC and **(B)** MDD patients. The maps reveal a widespread increase in stability within higher-order association regions, highlighting the consistent functional stability architecture observed across both HC and MDD groups.

Figure 3. Functional stability in MDD compared with HC. Brain maps of t-statistics show the significant clusters surviving the permutation test with threshold-free cluster enhancement (PT with TFCE) in the two-sample t-test. The cluster-wise threshold was set at two-tailed $p < 0.025$.

Figure 4. Association between functional stability and clinical manifestations of depression. (A)

Temporal stability peak locations were extracted for further analyses from the significant clusters identified

in the stability comparison between MDD vs. HC. **(B)** Partial correlation analyses between peak values of

functional stability and clinical manifestations of depression, including both the overall severity of the

disease and the severity of individual symptoms, as assessed by the HAMD 17-item scale. Total refers to

the total score of the HAMD 17-item scale, representing the overall severity of depressive symptoms. **(C)**

The scatter plot shows the correlation between item scores and Z-standardized stability values extracted

from the corresponding sensitive brain region (with $p < 0.05$) in MDD patients. Note that some distinct brain

regions are sensitive to the same symptom, along with some symptoms associate with the same brain

region. From these results, we identified a set of symptom-associated regions of interest including SFG (L

& R), INS, Sup. (R), and PCG (R) associated with insomnia, feelings of guilt, and insight symptoms. The

abbreviations represent the following: L (left hemisphere), R (right hemisphere), PCG (postcentral gyrus),

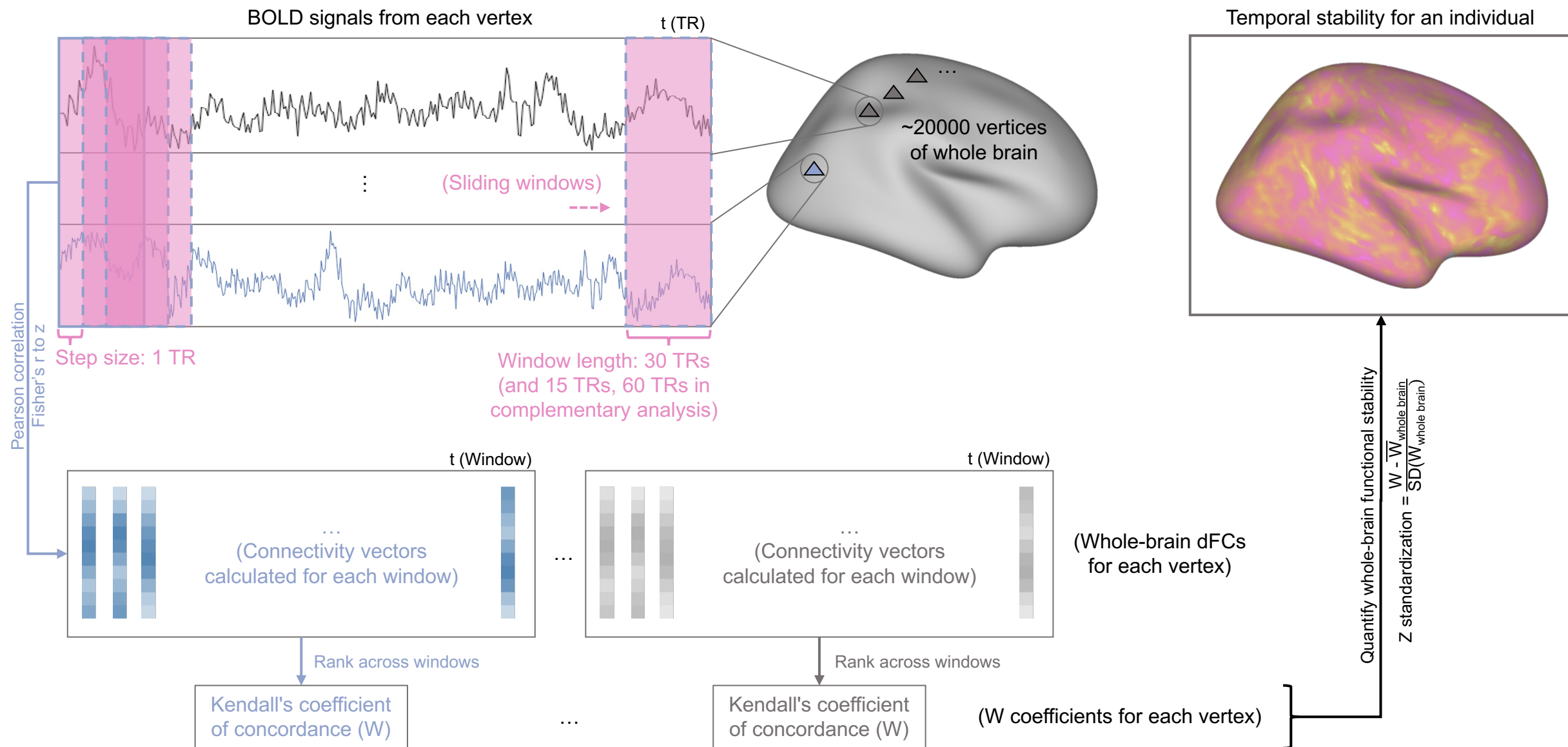
OP (occipital pole), INS, Sup. (superior insula sulcus), SOG (superior occipital gyrus), ITG (inferior temporal

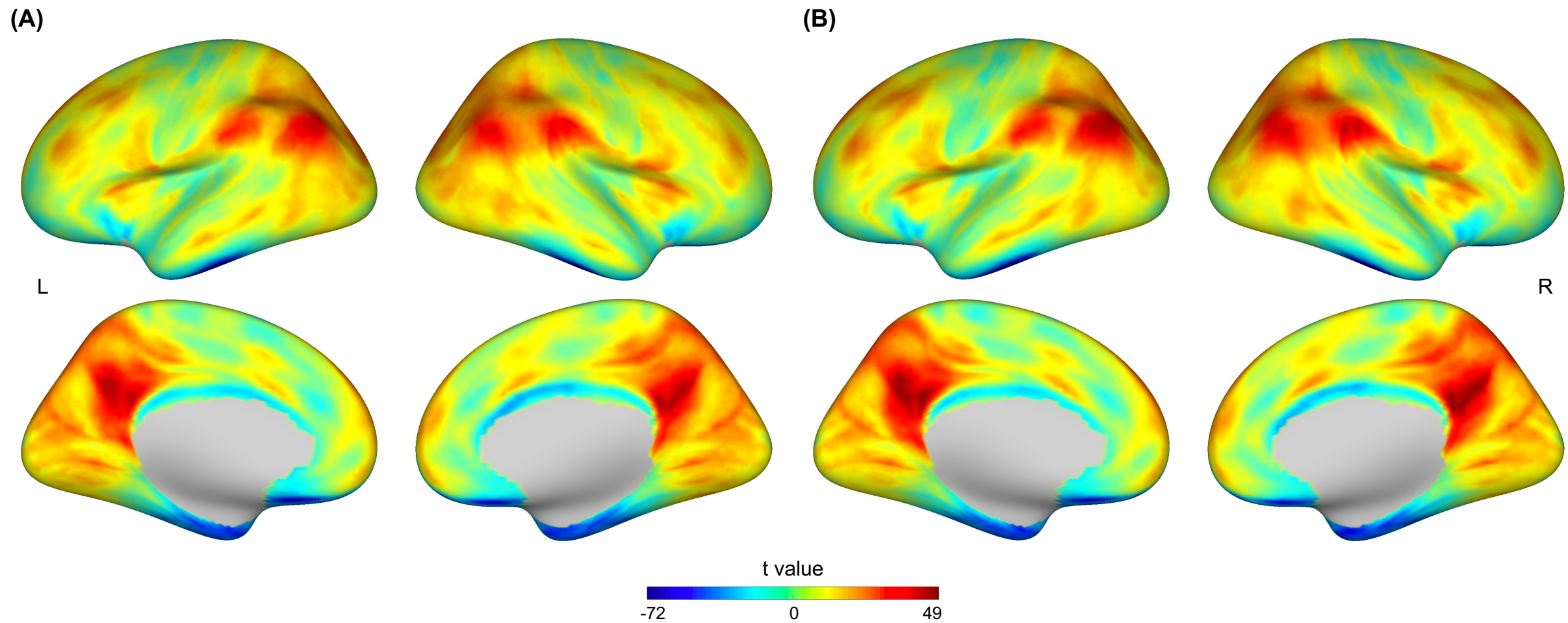
gyrus), TTG (temporal transverse gyrus), MOG (middle occipital gyrus), SFS (superior frontal sulcus), SFG

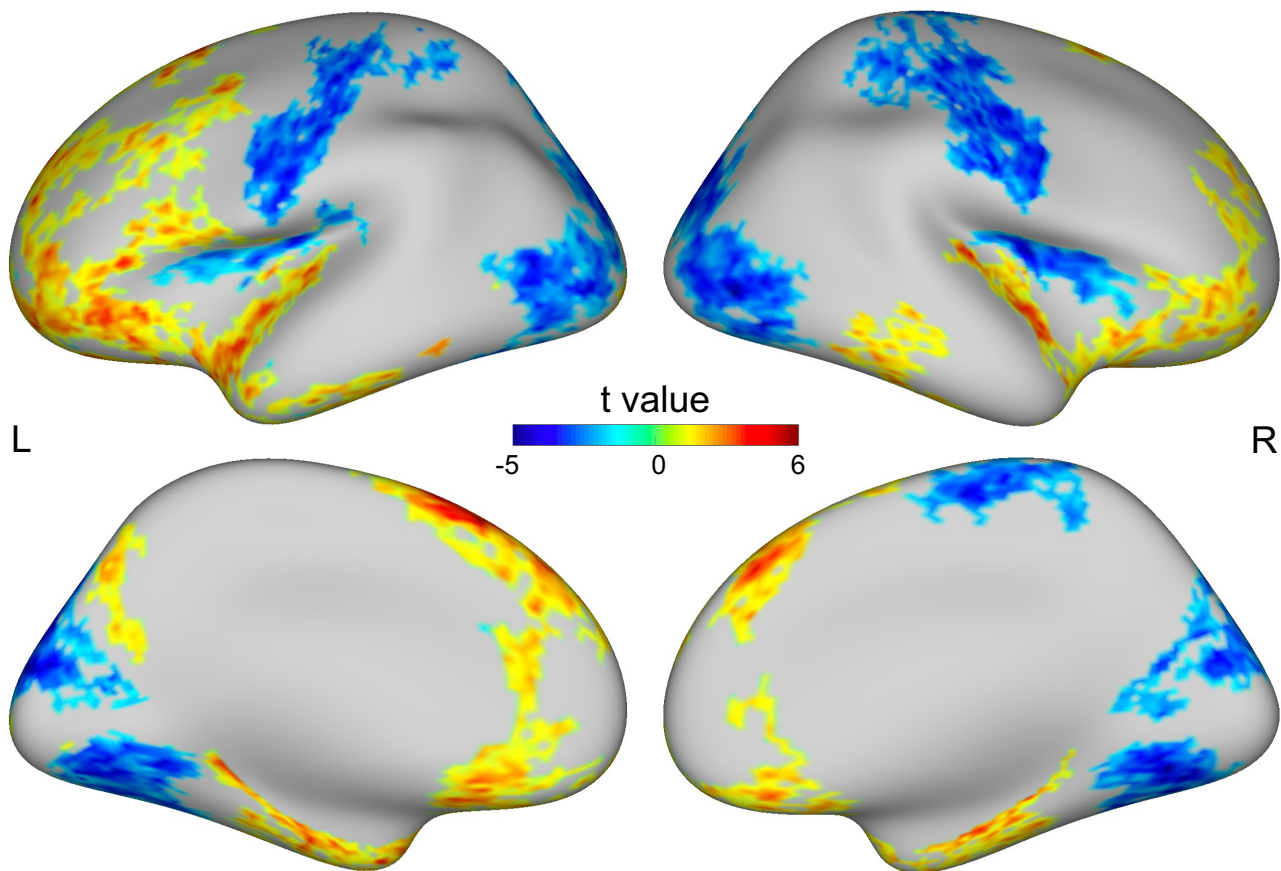
(superior frontal gyrus), D (depressed mood), FG (feelings of guilt), S (Suicide), IE (insomnia - early), IM

(insomnia - middle), IL (insomnia - late), WA (work and activities), PR (psychomotor retardation), PA (psychomotor agitation), AP (anxiety psychic), AS (anxiety somatic), GISS (gastrointestinal somatic symptoms), GNSS (general somatic symptoms), GS (genital symptoms), H (Hypochondriasis), WL (Weight loss), I (Insight). **: $p < 0.01$, *: $p < 0.05$.

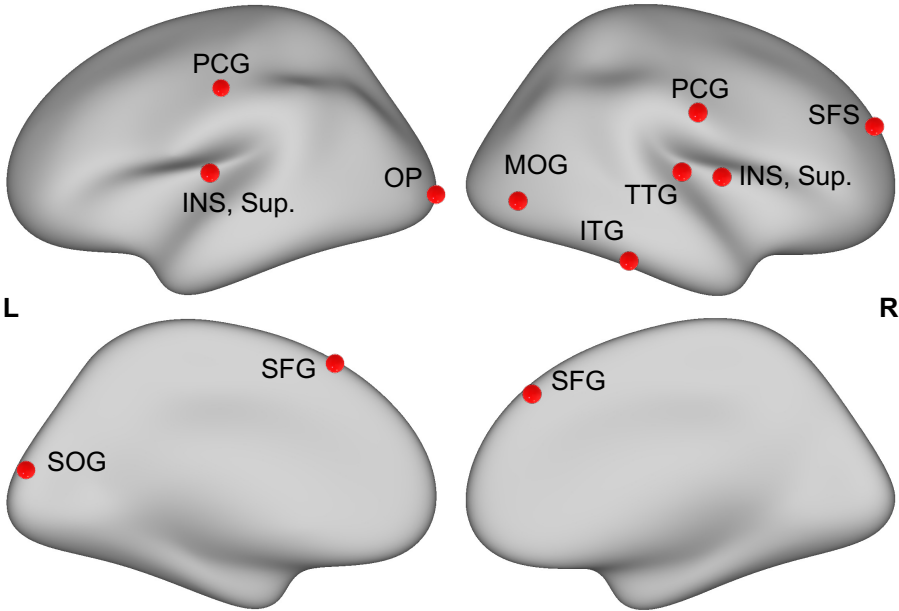
Figure 5. Dynamic functional connectivity in MDD compared with HC. Brain maps of t-statistics show the significant clusters of seed-based dynamic functional connectivity variation **(A)** between the left SFG and the whole brain, **(B)** between the right SFG and the whole brain, **(C)** between the right PCG and the whole brain, and **(D)** between the right INS, Sup. and the whole brain. These clusters survived PT with TFCE in the two-sample t-test, with the cluster-wise threshold set at $p < 0.05$. The distributed patterns of the significant clusters and their corresponding symptom-associated regions of interest showed notable overlap, suggesting the potentially shared aberrant dynamic network associated with insomnia, feelings of guilt, and insight symptoms in MDD. The abbreviations represent the following: SFG (superior frontal gyrus), PCG (postcentral gyrus), INS, Sup. (superior insular sulcus).



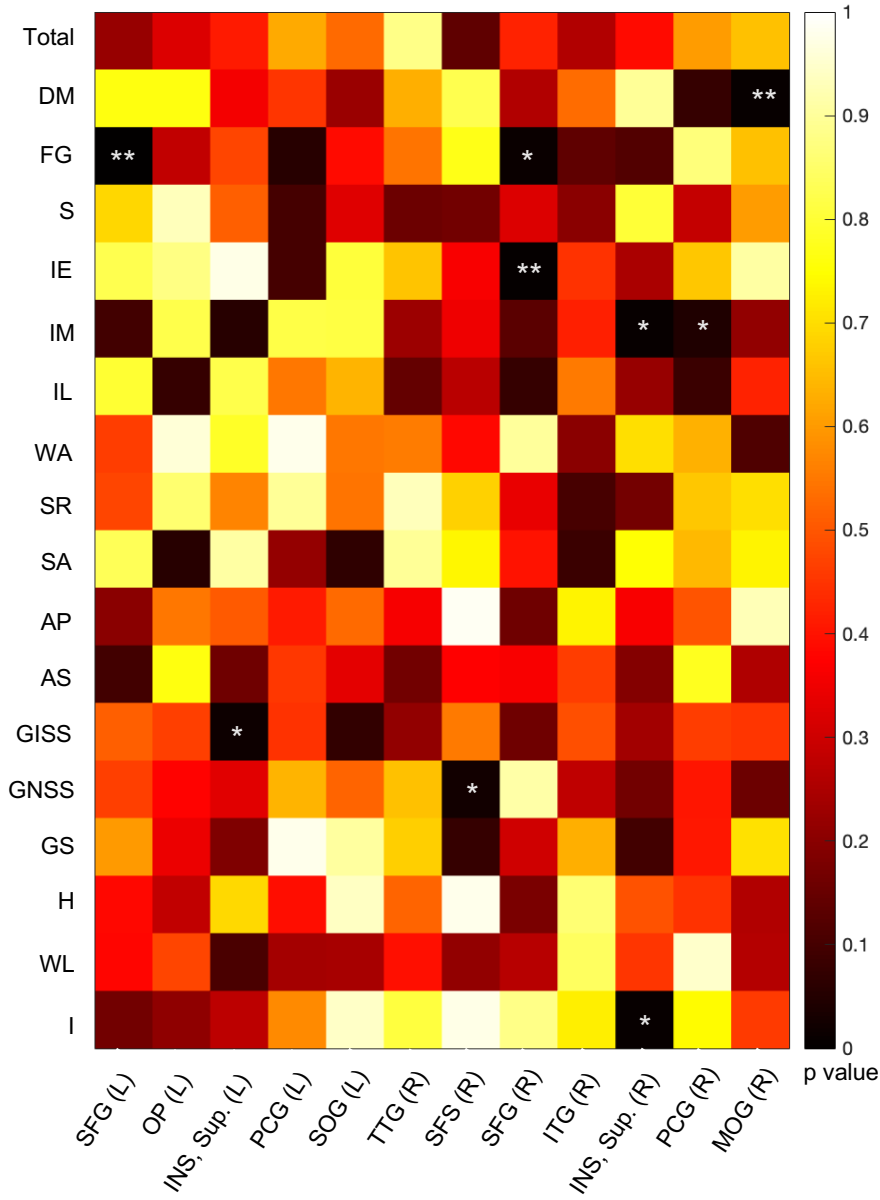




(A)



(B)



(C)

